

Unit: III
Physical Devices
Lecture: 1
Characteristic of Storage Devices

A storage device is used in the computers to store the data. There are four type of storage:

- Primary Storage
- Secondary Storage
- Tertiary Storage
- Off-line Storage

Some of the characteristics/ features of storage devices are:

- Volatility
- Accessibility
- Mutability
- Addressability

Volatility: There are two types of volatility:

- **Volatile Memory:**
 - Requires constant power to maintain the stored information.
 - All contents are erased when the system's power is turned off or interrupted.
 - Also known as temporary memory.
- **Non-Volatile Memory:**
 - will retain the stored information even if it is not constantly supplied with electric power.
 - Non volatile memory is the device which keeps the data even when the current is off.
 - It is suitable for long-term storage of information.

Accessibility: It refers to reading or writing of data records.

There are two types of accessibility:

- **Random Access:**
 - Any location in storage can be accessed at any moment in approximately the same amount of time.
 - Such characteristic is well suited for primary and secondary storage.

- **Sequential Access:**

- The accessing of pieces of information will be in a serial order, one after the other; therefore the time to access a particular piece of information depends upon which piece of information was last accessed.
- Such characteristic is typical of off-line storage.

Mutability: Mutability refers to the characteristic that **allows information to be overwritten at any time.**

There are three types of mutability:

- **Read/write storage or mutable storage:**

- Allows information to be overwritten at any time.

- **Read only storage:**

- Retains the information stored at the time of manufacture and write once storage (WORM) i.e. allows the information to be written only once at some point after manufacture.
- These are called immutable storage.

- **Slow write, fast read storage:**

- allows information to be overwritten multiple times but with the write operation being much slower than the read operation.

Addressability: There are three types of addressability:

- **Location-addressable:**

- Each individually accessible unit of information in storage is selected with its numerical memory address.

- **File addressable:**

- Information is divided into files of variable length, and a particular file is selected with human-readable directory and file names.

- **Content addressable:**

- Each individually accessible unit of information is selected based on the basis of the contents stored there.
- Content-addressable storage can be implemented using software (computer program) or hardware (computer device) with hardware being faster but more expensive option.
- Hardware content addressable memory is often used in a computer's CPU cache.